BHSM Galaxy Envelope Formation and Persistence

Conditional mathematical development of Norman Carberry’s collective rotation hypothesis

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Mathematical development prepared with ChatGPT
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Main result. A proposed envelope state can be connected to an explicit family of exterior gravitational fields using the earlier BHSM galaxy flux assumptions. A separate amplitude model can exhibit different formation and persistence limits. Neither construction yet derives the physical rotation trigger, its coefficients, or a galaxy lifetime from the BHSM action.

The concrete advance is a calculable structure with several independent checks: collective motion is represented by mass current and kinetic stress; rotation can favor or resist a boundary deformation depending on its inertia; persistence requires a stable state and a dynamical mechanism; and a specified source amplitude produces a definite radial acceleration profile.

Result obtained	Scientific status
Kinetic stress survives zero net rotation	Exact kinematic identity for the specified phase-space distribution.
Rotational contribution to boundary stiffness	Exact in the stated fixed angular momentum mechanical reduction; the BHSM mode and inertia are not supplied.
Separate formation and persistence limits	Exact for an illustrative sextic potential; its symmetry and coefficients are new assumptions.
Envelope amplitude to apparent halo profile	Exact conditional integration of the inherited spherical canonical-flux model.
Prediction for Draco or AGC 114905	Not evaluated. No observational fit or BHSM numerical solution is claimed.

The source record consists of the September galaxy notes and canonical envelope statement [S1-S3]. Their threshold, source-export identification, full field/domain solution, and metric slip remain assumptions or open derivations in those notes. This development does not promote those assumptions to established physics. It also does not change any existing BHSM completion status.

1 Collective motion with a well defined physical meaning

Work in the galaxy center-of-mass frame, within a stated aperture and averaging scale. Let f(x,v,t) denote mass per phase-space volume. The mass density, mean flow, and velocity-dispersion stress are

$$\rho = \int f d^3v, \qquad u_i = \rho^{-1} \int v_i f d^3v$$

$$\Pi_{ij} = \int (v_i - u_i)(v_j - u_j) f d^3v$$

$$K_{ij} = \rho u_i u_j + \Pi_{ij}, \quad T_{\mathrm{kin}} = \tfrac{1}{2} \int K_{ii}\, d^3x$$

$$\mathbf{J} = \int \mathbf{x} \times (\rho \mathbf{u})\, d^3x$$

K is the spatial kinetic stress. These quantities do not depend on how a fixed amount of matter is split into sample particles. Uniform translation has been removed. The division into streaming and dispersion depends on averaging, so a consistent total second moment is essential. Field stresses, gas thermodynamics, boundary fluxes, and gravitational terms must enter through the actual action as well.

A direct counterrotation calculation

Take equal collisionless streams with velocities +v0 and -v0 around the same axis, at the same positions. Their local mean flow and net angular momentum vanish. Their azimuthal stress and kinetic energy do not:

$$\mathbf{u} = 0, \quad \mathbf{J} = 0, \quad K_{\phi\phi} = \rho v_0^2, \quad T_{\mathrm{kin}} = \tfrac{1}{2} \int \rho v_0^2\, d^3x > 0$$

A single corotating stream of the same density and speed has the same spatial kinetic stress, but a nonzero mass current. A mechanism depending only on K cannot distinguish these instantaneous states. A coupling to current, orbital organization, or history might distinguish them. The full relativistic stress-energy tensors differ in their momentum density; this is not an equivalence of the full states.

What can be measured before an encapsulation law is known

Angular-momentum alignment, organized kinetic-energy fraction, the dispersion tensor, and their persistence are candidate diagnostics. None is automatically the BHSM control parameter. In particular, a low net spin does not establish that a galaxy is dynamically still, or that it lacks the proposed envelope.

2 How rotation changes a boundary deformation

Let q be a physical deformation coordinate and I(q) its axial moment of inertia. At fixed angular momentum J, the minimum organized rotational energy is J squared divided by twice I. Differential motion and velocity dispersion must remain in the residual energy. For an axial decomposition this statement is exact:

$$T_{\mathrm{stream}} = \frac{J_z^2}{2I} + \frac{1}{2}\int \rho |\mathbf{u}_{\mathrm{res}}|^2 \, d^3x$$

The residual has zero angular momentum along the selected axis. This identity does not mean that a collisionless galaxy can relax to rigid rotation. Separate stream momenta, orbital actions, and other conserved phase-space information may prevent that relaxation.

Fixed angular momentum

$$\mathcal{E}_J(q) = U(q) + \frac{J^2}{2I(q)}$$

$$\frac{d^2\mathcal{E}_{J,\mathrm{rot}}}{dq^2} = \frac{J^2(I')^2}{I^3} - \frac{J^2 I''}{2I^2}$$

This contribution has no universal sign. For homologous size R with I = k M R squared, the rotational curvature is positive:

$$\frac{d^2\mathcal{E}_{J,\mathrm{rot}}}{dR^2} = \frac{3J^2}{kMR^4} > 0$$

At fixed J, contraction raises the rotational energy. Conversely, for the illustrative shape dependence I(q) = I0(1 + q squared), the rotational curvature at zero is -J squared/I0. Rotation favors that shape deformation to quadratic order. The actual BHSM mode must determine which dependence applies.

A reservoir holding angular velocity fixed gives another problem

$$\mathcal{F}_\Omega = \mathcal{E} - \Omega J = U(q) - \frac{1}{2} I(q)\Omega^2$$

$$\mathcal{E}''_{J,\mathrm{rot}} - \mathcal{F}''_{\Omega,\mathrm{rot}} = \Omega^2 \frac{(I')^2}{I} \geq 0$$

The comparison uses J = I Omega at the same instantaneous state. The negative rotational term in the fixed-Omega potential includes work by the reservoir. Holding the external BHSM environment fixed does not, by itself, select either constraint: interface torques can still exchange angular momentum. A fixed-J approximation requires an appropriate angular-flux condition or timescale separation.

3 The physical reduction that BHSM must supply

The rotation calculation identifies legitimate inputs to a reduction. It does not supply the galaxy-scale BHSM action, physical mode, normalization, or admissible constraints. Let y denote the other internal and interface variables. A conditional reduced stationary energy can be written as

$$\mathcal{E}_{\mathrm{red}}(q;\Xi,\mathscr{I},e)=\mathcal{E}(q,y_*(q);\Xi,\mathscr{I},e)$$

Here e is the prescribed parent/environment data, held fixed in the variation; the interface variables adjust through the action equations. The set of retained conserved data is denoted by the script I. It can include more than total angular momentum. Across different galaxies the parent data may differ.

$$H_{\mathrm{red}}=H_{qq}-H_{qy}H_{yy}^{-1}H_{yq}$$

This Schur expression applies only after gauge and constraint directions are handled and the indicated complement is invertible on its physical domain. For a true stability conclusion the full constrained problem, its kinetic structure, and other modes remain relevant. A scalar local minimum can still be unstable in a direction omitted from the reduction.

A fixed physical normalization matters

Choose a dimensionless q using a declared boundary mode and positive kinetic normalization. Let E0 be a fixed positive energy scale. The dimensionless quadratic coefficient is then the physical reduced curvature divided by E0. The quartic and higher coefficients must be obtained from the same reduction, including the response of y. They cannot be imported from an unrelated microscopic branch.

$$\alpha=E_0^{-1}\partial_q^2\mathcal{E}_{\mathrm{red}}(0)$$

Dimensionless descriptors such as the following can organize a calculation, without asserting that any one selects the event:

$$\Xi_b=\frac{GM_bR_H}{c^2R^2},\qquad \chi_J=\frac{J^2}{GM_b^3R}$$

$$\chi_\sigma=\frac{2T_{\mathrm{disp}}R}{GM_b^2},\qquad \varepsilon_H=\frac{R}{R_H}$$

The imported galaxy notes use the baryonic curvature control Xi and leave the critical root xi_c unresolved [S1,S2]. Adding rotation can change the critical state, move or deform the seam, and change the baryonic distribution. Those changes must be solved together rather than assigning an extra centrifugal source on top of an unchanged equilibrium.

A vanishing physical Hessian eigenvalue identifies a possible marginal point. Finite encapsulation still requires an admissible trajectory and completed boundary, field, and domain matching. A first-order transition can also occur across a barrier while the reference-state curvature remains positive.

4 Formation and persistence are distinct mathematical questions

To test the logic of a persistent state, consider a reduced amplitude potential. The following polynomial is an illustrative normal form, not a BHSM-derived energy. The amplitude is dimensionless and E0 carries energy units; a, b, and c below are dimensionless.

A positive quartic does not by itself provide hysteresis

$$\frac{\mathcal{E}_4}{E_0} = \frac{a}{2}q^2 + \frac{b}{4}q^4, \qquad b > 0$$

The reference state q = 0 is stable for a greater than zero. For a less than zero, the stable amplitudes satisfy q squared = -a/b. The amplitude returns continuously to zero at the same threshold when the control reverses. There is no range where the reference and nonzero states are both local minima.

A negative quartic and positive sextic permit coexistence

$$\frac{\mathcal{E}_6}{E_0} = \frac{a}{2}q^2 - \frac{b}{4}q^4 + \frac{c}{6}q^6, \qquad b, c > 0$$

$$\partial_q \mathcal{E}_6 = E_0 q(a - bq^2 + cq^4)$$

$$y_\pm = q_\pm^2 = \frac{b \pm d}{2c}, \qquad d = \sqrt{b^2 - 4ac}$$

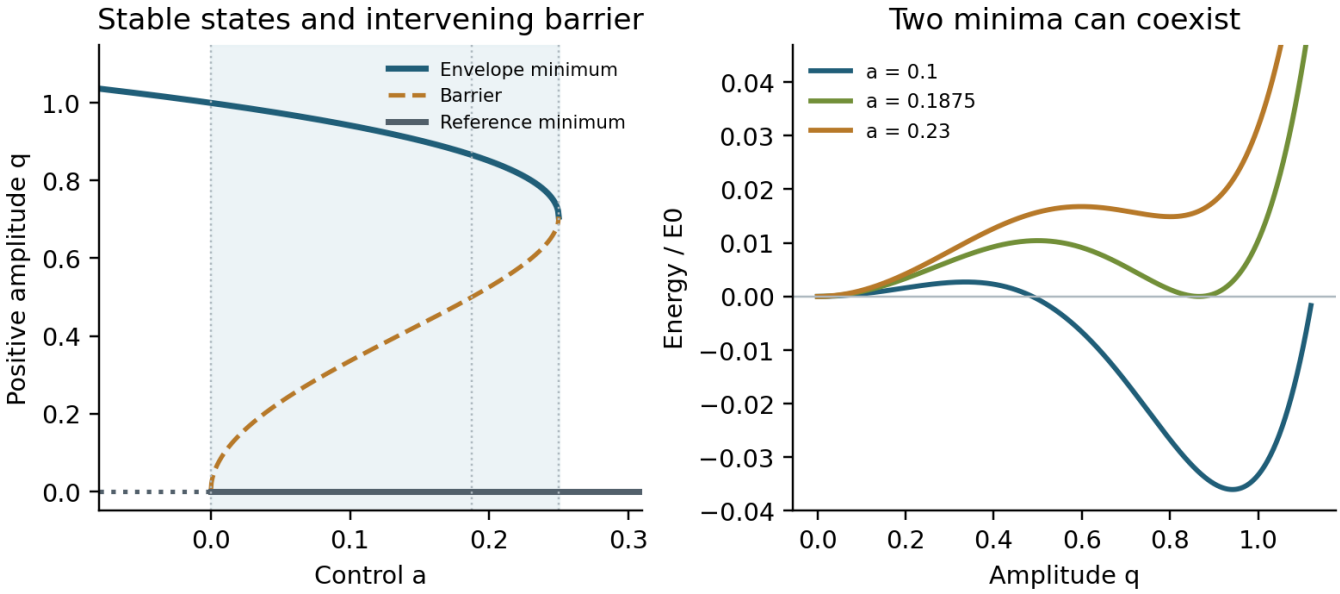
Where both roots are positive, y+ gives a nonzero minimum and y- gives the intervening barrier. Their curvatures are +2 E0 y+ d and -2 E0 y- d respectively. The reference-state curvature is E0 a.

Boundary	Value of a	Meaning
Reference loses stability	0	For a below zero the reference state is unstable.
Equal energy	3 b squared / (16 c)	The reference and nonzero minima have the same energy.
Nonzero branch ends	b squared / (4 c)	The nonzero minimum and barrier merge; the minimum ceases to exist.

Both minima exist for 0 < a < b squared/(4c). Below the equal-energy value the envelope minimum has lower energy; above it the envelope is metastable. At either loss-of-stability boundary the limiting point is not a strict stable quadratic minimum.

An even potential assumes an appropriate q to -q symmetry. That symmetry has not been established for the galactic BHSM mode. Allowed linear or cubic terms change these formulas; a linear source generally removes q = 0 as an exact stationary state.

5 What the persistence model shows



Illustrative dimensionless normal form with b = c = 1. Solid branches are local minima, the dashed nonzero branch is the barrier, and the shaded interval contains two minima. Negative-amplitude copies are omitted. No galaxy parameters are assigned.

At the same present control a, a system can occupy the reference minimum or the envelope minimum. The previous trajectory can therefore matter if the dynamics retains a metastable state. That is the mathematical ingredient required for a past encapsulation event to survive after the original driving condition weakens.

A concrete normalized example

At a = 3/16 and b = c = 1, the reference minimum is q = 0, the barrier is q = 1/2, and the nonzero minimum is q = square root of 3 divided by 2. The two minima have equal energy and each crossing barrier has height E0/96. These numbers demonstrate the mechanism in the specified polynomial; they are not a galactic threshold or an energy estimate.

Two minima permit hysteresis but do not guarantee it

Starting in the reference state and decreasing a, weakly perturbed dynamics can retain that local minimum until a approaches zero. After the nonzero state forms, increasing a can retain it up to the branch endpoint at b squared/(4c). A system that always reaches its lowest-energy state instead switches at the equal-energy value. Noise, nucleation, driving rate, and alternative transition paths determine the actual switching point.

In exactly symmetric deterministic dynamics, a perfectly exact q = 0 trajectory can remain at zero after becoming unstable; a perturbation is needed to depart. The loss of stability is therefore a statement about sensitivity to perturbations, not a proof that the finite event has occurred.

If BHSM produces this bistability, a suitable earlier history could leave an envelope after organized rotation decreases. Draco remains a test: this illustration has not assigned it a past transition or placed it inside a physical persistence interval.

6 Energy barriers do not determine a galaxy lifetime

The barrier heights can be calculated exactly within the sextic normal form. In the coexistence interval define d = square root of (b squared - 4ac). The forward barrier from the reference state and the reverse barrier from the envelope are

$$\Delta\mathcal{E}_{\mathrm{form}} = E_0 \frac{(b-d)^2(b+2d)}{48c^2}$$

$$\Delta\mathcal{E}_{\mathrm{loss}} = E_0 \frac{d^3}{12c^2}$$

At equal energy they are both E0 b cubed/(96 c squared). The formation barrier vanishes as a approaches zero from above. The reverse barrier vanishes as the envelope branch reaches its endpoint. These are barriers along one amplitude coordinate; other modes may provide lower-energy escape paths.

One illustrative dissipative equation

$$\dot{q} = -\Gamma\, \partial_q \mathcal{E}, \qquad \Gamma > 0$$

$$\dot{\mathcal{E}} = -\Gamma (\partial_q \mathcal{E})^2 \le 0$$

The energy inequality holds at fixed control. Gamma has units of inverse energy times inverse time for dimensionless q. Near a stable minimum q*, the linear relaxation time is

$$\tau_{\mathrm{relax}} = \frac{1}{\Gamma \mathcal{E}''(q_*)}$$

Neither Gamma nor an overdamped equation has been derived from BHSM here. A conservative reduction instead has inertia; radiation, phase mixing, other fields, boundary exchange, and memory may supply a different effective evolution. With a time-dependent control, the energy balance also includes work from the changing control.

What persistence would actually require

A physical persistence result requires the full stable configuration, its allowed perturbations, the relevant kinetic law, and environmental coupling. A potential with two minima supplies no duration in years. Thermal escape formulas cannot simply be applied to a galaxy without deriving the fluctuation statistics and coupling that make them appropriate.

Accordingly, this note establishes a possible mathematical separation between formation and loss conditions. It supplies neither a physical formation time nor survival for billions of years. Those quantities remain outputs of the actual BHSM evolution problem.

7 A conditional gravitational law for the envelope amplitude

The earlier galaxy solution [S1,S2] proposed a canonical scalar flux and a spherical exterior source. Preserve its assumptions explicitly. Let Mb be the approximately enclosed baryonic mass outside a seam radius r*, and set

$$W = V_\Sigma^2 = \frac{GM_b}{r_\star}, \qquad a_\star = \frac{GM_b}{r_\star^2} = \xi_c \frac{c^2}{R_H}$$

Introduce a dimensionless relative export amplitude A. It is the proposed source strength relative to the earlier branch. This is new bookkeeping for an unknown physical export law; it is not a measured dark-matter fraction. Assume it is nonnegative and constant across the exterior region.

$$QS_{\mathrm{enc}}(r) = \frac{AW}{r^2}, \qquad r \geq r_\star$$

Let g_enc denote the positive inward magnitude of the additional acceleration. With continuous activation at the seam, the canonical-flux equation gives

$$\frac{d}{dr}(r^2 g_{\mathrm{enc}}) = AW, \qquad g_{\mathrm{enc}}(r_\star) = 0$$

$$g_{\mathrm{enc}} = AW\left(\frac{1}{r} - \frac{r_\star}{r^2}\right)$$

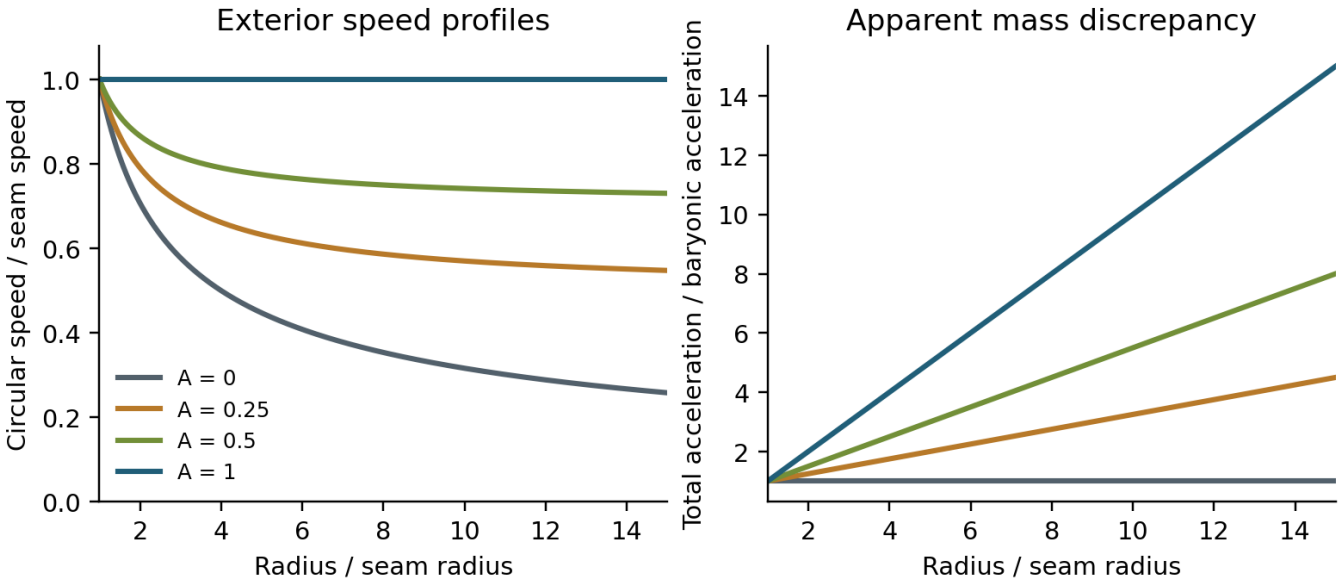
Adding the exterior baryonic field once yields the main conditional prediction:

$$v_c^2(r) = A\frac{GM_b}{r_\star} + (1 - A)\frac{GM_b}{r}$$

Relative amplitude	Exterior behavior
A = 0	Baryonic Keplerian decline.
0 < A < 1	Decline toward a nonzero plateau if enough radial range exists.
A = 1	Exactly flat in this idealized spherical exterior.
A > 1	Rise toward a higher plateau; the effective added mass is still positive.

The derived integration does not establish the assumed source export, the scalar field/domain solution, or the metric response. Rotation can also move and deform the seam. Those effects are held fixed in this conditional family and must be restored in the physical calculation.

8 Observable consequences of a specified export amplitude



Conditional profiles for prescribed amplitudes. The seam speed is square root of (G Mb/r*). The same ordinary-matter mass and seam are used for all curves. These are theoretical illustrations, not observations or fitted galaxy curves.

The acceleration relation and discrepancy follow directly, for the spherical exterior where gb is at most a*:

$$g_{\mathrm{tot}} = (1-A)g_b + A\sqrt{a_\star g_b}, \qquad g_b = \frac{GM_b}{r^2}$$

$$D(r) = \frac{g_{\mathrm{tot}}}{g_b} = 1 + A\left(\frac{r}{r_\star} - 1\right)$$

$$V^4_{\mathrm{plateau}} = A^2 \xi_c G M_b \frac{c^2}{R_H}$$

The earlier mass-speed result is recovered at A = 1. If the physical state changes A or the seam threshold xi_c across galaxies, the mass-speed normalization constrains the product A squared times xi_c. A freely fitted amplitude for every galaxy would weaken predictive power; the action and state must determine it.

The effective halo is another representation of the same field

$$M^{\mathrm{eff}}_{\mathrm{enc}} = \frac{AM_b}{r_\star}(r - r_\star), \qquad \rho^{\mathrm{eff}}_{\mathrm{enc}} = \frac{AM_b}{4\pi r_\star r^2}$$

This density is what a Newtonian reconstruction would assign to the proposed extra field. It is not a second independent mass component to add to that field. The extra acceleration and enclosed effective mass start at zero at the seam. A finite-thickness boundary would smooth the abrupt density activation.

9 Radial variation and the finite cosmic environment

A physical envelope need not have a constant export amplitude. For a specified radial function A(r), integration instead gives

$$g_{\mathrm{enc}}(r) = \frac{W}{r^2}\int_{r_\star}^{r} A(s)\,ds$$

$$v_c^2(r) = \frac{GM_b}{r} + \frac{W}{r}\int_{r_\star}^{r} A(s)\,ds$$

$$\rho_{\mathrm{enc}}^{\mathrm{eff}}(r) = \frac{WA(r)}{4\pi G r^2}$$

These formulas make the missing physics explicit. The action must predict A(r), rather than reconstruct an arbitrary function from the rotation curve being explained. A local amplitude q does not automatically imply A = q squared. Even a symmetry allowing a leading quadratic term leaves its coefficient and normalization to be derived.

A plateau is local rather than an infinite halo

If a constant positive source ends at an outer radius r_cut, its exterior contribution becomes

$$g_{\mathrm{enc}}(r) = \frac{AW(r_{\mathrm{cut}} - r_\star)}{r^2}, \qquad r > r_{\mathrm{cut}}$$

The circular-speed curve ultimately declines again. For a small A, the approach to a plateau requires r much larger than |1-A| r*/A and may not occur before the source ends. V_plateau denotes the local intermediate behavior or the formal untruncated approximation, not a physical speed at infinite radius.

A compact hypersphere requires global source balance

For a smooth scalar field with the stated operator on a compact boundaryless S3, integration of Laplacians gives

$$\int_{S^3} (\Delta T - B\Delta^2 T)\,dV = 0$$

$$\int_{S^3} S_{\mathrm{total}}\,dV = 0$$

A positive local source therefore needs compensating source, return flow, or an explicitly different global domain or zero-mode equation. Cutting the source off makes its integrated strength finite but does not provide that compensation. The local attractive branch is not a complete global solution by itself.

10 Tests with rotating and pressure supported galaxies

The first empirical test is not whether a toy potential can have two minima. It is whether the physical action predicts the source and the branch occupation for systems whose gravity is measured independently. Present motion and any reconstructed history must be used without defining an envelope by the mass discrepancy being explained.

Draco tests a strong present rotation requirement

Vitral and collaborators combined Hubble proper motions and line-of-sight velocities to infer a dark component in Draco while finding modest ordered rotation, with mean v/sigma = 0.22 +/- 0.09 [O1]. A rule requiring strong present organized rotation would have difficulty with this system. The present note has not shown that Draco formed an earlier envelope or that its state is metastable.

Pressure-supported galaxies require a dynamical prediction for dispersions. In an idealized spherical equilibrium, the relevant relation is the Jeans equation:

$$\frac{d(\nu\sigma_r^2)}{dr} + \frac{2\beta\nu\sigma_r^2}{r} = -\nu g_{\mathrm{tot}}$$

Nu is tracer density and beta is the velocity-anisotropy parameter. Predictions require tracer profiles, anisotropy, projection, and observational selection. This equation is not an adequate replacement for the axisymmetric analysis used in the cited Draco work. A circular speed inferred from a potential is not the observed mean stellar rotation.

AGC 114905 tests organization without a large inferred halo

Deep imaging finds a disc with spiral-arm-like features and an inclination near 31 +/- 2 degrees, while the inferred halo has unusually low density [O2]. It should not be labeled securely dark-matter-free. Updated idealized simulations found that revised disc and low-density halo models could remain stable for five billion years [O3]. That stability result belongs to those conventional models, not to BHSM.

Population constraints and lensing

The observed radial-acceleration relation connects baryonic and dynamical acceleration across varied disc galaxies [O4]. The physical dependence of A squared times xi_c must preserve that relation or predict its departures. Distance, inclination, mass-to-light ratio, pressure corrections, and shared measurement errors must be included in any comparison.

Circular motions constrain the time potential Phi. Weak-field lensing depends on Phi + Psi. The source integration above does not determine Psi or the gravitational slip. Both potentials and finite-region matching must come from the same action, and the optical contribution must be counted once. A rotation fit alone does not complete this prediction.

11 What has been resolved and what must be calculated

Obtained in this development	Still required for a physical BHSM prediction
Definitions of current, kinetic stress, and angular momentum	The actual action coupling of these quantities to the galaxy boundary mode.
Fixed-J and fixed-Omega rotational stiffness formulas	The physical mode q, its inertia I(q), and correct conserved/interface data.
Exact bistability interval and barriers in a normal form	The symmetry, energy scale, coefficients, and stability of all remaining modes.
An explicit formation-versus-loss mechanism	An admissible finite formation history and a derived kinetic or memory law.
Exact conditional radial gravity for a given amplitude	The source export A(q, fields, history), seam location and xi_c, and global completion.
A map from extra field to an effective halo representation	The complete metric, slip, lensing, and quantitative tests against data.

Next physical object. Compute the constrained galaxy-envelope energy or action on a declared mode using the baryonic phase-space state, the existing BHSM terms, and the fixed parent data with admissible interface response. Evaluate its curvature and nonlinear coefficients, then derive the exported scalar charge from the same solution. This determines whether a rotating state actually favors encapsulation and whether a finite persistent branch exists.

The conditional radial map is usable immediately once an independently predicted source amplitude and seam are supplied. The normal-form calculation is useful for identifying what a persistence mechanism must contain. Neither substitutes for the missing physical map from collective motion and history to a boundary state and its gravitational source.

Verification and scope

The branch roots, equal-energy value, barrier differences, and radial identities were checked analytically and at representative normalized values. Independent reviews checked the mechanical constraints, nonlinear stability formulas, and source integration. These checks verify the stated mathematics. They do not certify the BHSM field equations or establish agreement with a galaxy dataset.

No new physical acceleration scale, fitted galaxy coefficient, rotation threshold, or lifetime has been measured or derived in this note. The physical coefficients remain unevaluated. The contribution is a conditional calculation with explicit tests and a precise set of remaining action outputs.

Sources and notation

Prior BHSM source record

[S1] BHSM_Galaxy_Gravity_Aether_Branch_v0_4.md. 15 September 2026. Conditional gravity-driven galaxy branch, normalized curvature control, exterior rotation law, and unresolved action outputs.

[S2] BHSM_Topographic_Galaxy_Encapsulation_Solution_v0_3.md. 15 September 2026. Canonical-flux candidate, seam charge, spherical exterior integration, conditional mass-speed normalization, and lensing-slip obligation.

[S3] BHSM_Canonical_Theory_Statement_v0_2.md. 16 September 2026. Norman P. Carberry. Distinguishes encapsulation as a process, envelope as the general state, and enclosure as a persistent closed subclass.

Primary observational and modeling sources

[O1] Vitral et al. (2024). HSTPROMO Internal Proper Motion Kinematics of Dwarf Spheroidal Galaxies I. Velocity Anisotropy and Dark Matter Cusp Slope of Draco. Astrophysical Journal 970, 1. arxiv.org/abs/2407.07769.

[O2] Mancera Pina et al. (2024). Exploring the nature of dark matter with the extreme galaxy AGC 114905. Astronomy and Astrophysics 689, A344. arxiv.org/abs/2404.06537.

[O3] Afruni et al. (2025). The long life of ultra diffuse galaxies inside low-density dark matter halos the case of AGC 114905. arxiv.org/abs/2502.08717.

[O4] McGaugh, Lelli and Schombert (2016). The Radial Acceleration Relation in Rotationally Supported Galaxies. Physical Review Letters 117, 201101. arxiv.org/abs/1609.05917.

Symbols that refer to different quantities

Symbol	Meaning
T in the scalar operator	The inherited topographic scalar field.
T_kin or T_disp	Kinetic energy or its dispersion contribution, not the scalar field.
a in the normal form	Dimensionless reduced curvature coefficient, not an acceleration.
a_star	Dimensional galaxy acceleration scale xi_c c squared/R_H.
A	Relative scalar-source export amplitude, not a probability or assumed fraction.
q	Hypothetical normalized envelope coordinate; its physical mode is still to be specified.
W	Seam speed squared, equal to G Mb/r_star under the inherited matching assumption.